

INFRATHRONE (COREOPS TRACK)

WEEK 1 ASSIGNMENT PACK (Linux + Kubernetes Fundamentals)

Theme: Building the Foundation of Outage Handling

This assignment is designed to push you slightly outside your comfort zone while keeping everything achievable.

You must submit screenshots, notes, and your debugging explanation for each task.

The goal is not perfection; the goal is understanding.



Scheduling & Resource Debugging

Objective:

Understand why pods get stuck in Pending, which is one of the most common outage triggers.

Tasks:

1. Create a pod requesting high resources:

```
apiVersion: v1
kind: Pod
metadata:
  name: heavy
spec:
  containers:
  - name: load
    image: nginx
    resources:
      requests:
        cpu: "4"
        memory: "2Gi"
```

Apply: `kubectl apply -f heavy.yaml`

2. Investigate:
 - `kubectl describe pod heavy`
 - `kubectl get nodes`
 - `kubectl describe node <node-name>`
3. Identify why it's not scheduled.
4. Fix by lowering requests: `cpu: "0.2", memory: "128Mi"`

Submission:

- Pod events screenshot
- Node resource screenshot
- 5-line explanation of the root cause



THE DEVOPS WAR ROOM